

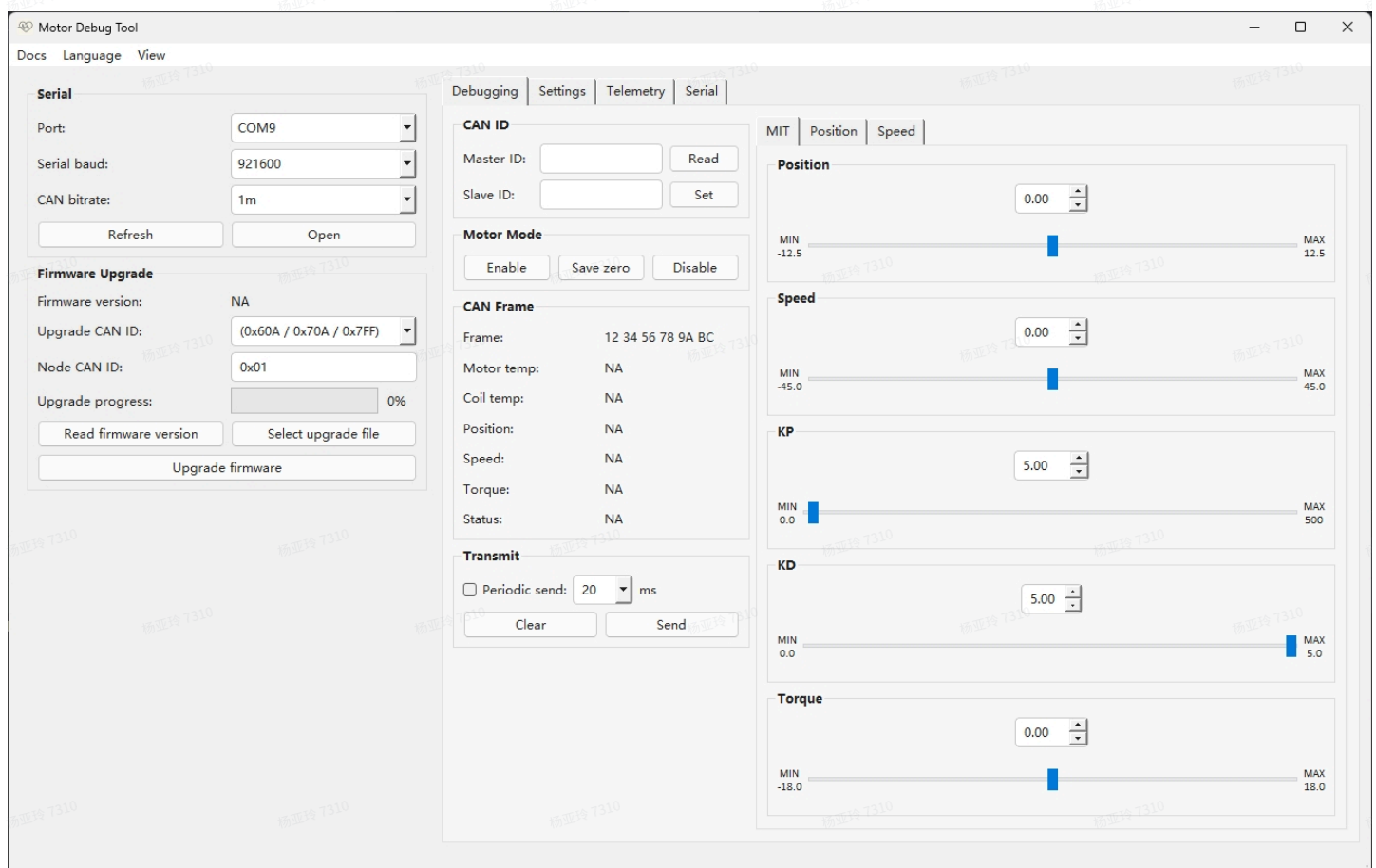
Debugging Tool Quick Guide

Main Interface Overview

https://github.com/OpenWonderLabs/RobotDoc/blob/main/tools/motor_debug_tool_v1.0.2.zip

1. Click the link above. Download the compressed the file, extract it, and run the .exe file in the folder.
 2. Use the Makerbase CANable V2.0 tool to connect to the PC, open the corresponding serial port, and begin functional debugging.
- Note: Before executing any debugging command (except firmware upgrades), you must first set or read the device's CAN ID.

1.1 Menu Bar



- Operation logs from the host PC software are saved in the `logs` folder.
- Language can be switched between Chinese and English.
- The "Telemetry" page shows data only when you set periodic sending.

1.2 Serial Port Section

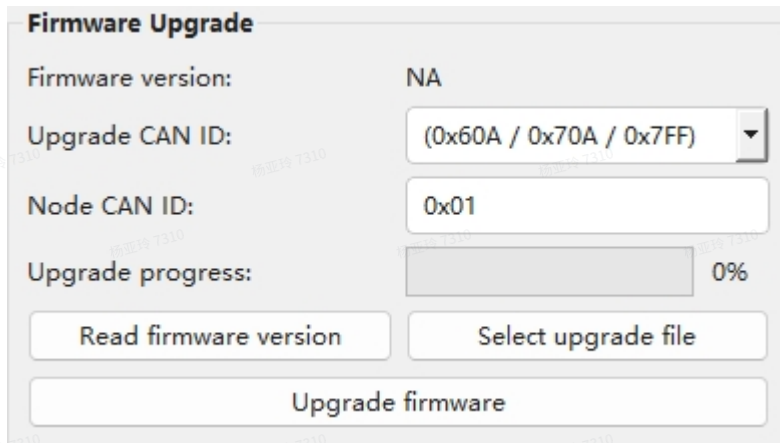


The Serial Port Configuration dialog box contains the following fields and buttons:

- Port:** A dropdown menu currently showing "COM9".
- Serial baud:** A dropdown menu currently showing "921600".
- CAN bitrate:** A dropdown menu currently showing "1m".
- Buttons:** "Refresh" and "Open".

- Supports refreshing, opening, and closing the serial port.
- The current motor only supports 1Mbps CAN baud rate. Use the default configuration.

1.3 Firmware Upgrade Section



The Firmware Upgrade dialog box contains the following fields and buttons:

- Firmware version:** A text field showing "NA".
- Upgrade CAN ID:** A dropdown menu showing "(0x60A / 0x70A / 0x7FF)".
- Node CAN ID:** A text field showing "0x01".
- Upgrade progress:** A progress bar showing "0%".
- Buttons:** "Read firmware version", "Select upgrade file", and "Upgrade firmware".

- Supports reading firmware version, selecting a file, and upgrading firmware.
- The upgrade target CAN ID is currently fixed by default.
- The node (device) CAN ID can be entered manually.

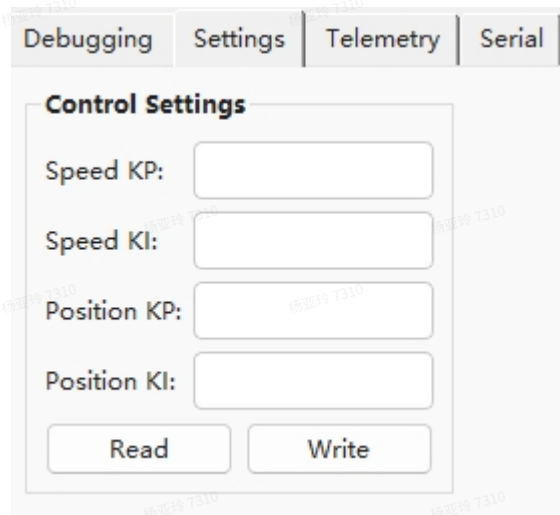
1.4 Debugging Page

The screenshot displays the 'Debugging' tab of a control interface. It is divided into two main columns. The left column contains the 'CAN ID' section with input fields for Master ID and Slave ID, and buttons for 'Read' and 'Set'. Below this is the 'Motor Mode' section with 'Enable', 'Save zero', and 'Disable' buttons. The 'CAN Frame' section shows a table of data: Frame (12 34 56 78 9A BC), Motor temp (NA), Coil temp (NA), Position (NA), Speed (NA), Torque (NA), and Status (NA). The 'Transmit' section has a 'Periodic send' checkbox set to 20 ms, with 'Clear' and 'Send' buttons. The right column features a sub-header with 'MIT', 'Position', and 'Speed' tabs. The 'Position' tab is active, showing a slider from -12.5 to 12.5 with a value of 0.00. Below it is the 'Speed' section with a slider from -45.0 to 45.0 and a value of 0.00. The 'KP' section has a slider from 0.0 to 500 with a value of 5.00. The 'KD' section has a slider from 0.0 to 5.0 with a value of 5.00. The 'Torque' section has a slider from -18.0 to 18.0 with a value of 0.00.

Section	Parameter	Value
CAN ID	Master ID	
	Slave ID	
Motor Mode	Enable	Button
	Save zero	Button
	Disable	Button
CAN Frame	Frame	12 34 56 78 9A BC
	Motor temp	NA
	Coil temp	NA
	Position	NA
	Speed	NA
	Torque	NA
	Status	NA
Transmit	Periodic send	20 ms
	Clear / Send	Buttons
Control Mode (Position Tab)	Position	0.00
	Speed	0.00
	KP	5.00
	KD	5.00
	Torque	0.00
	Slider Ranges	Position: -12.5 to 12.5; Speed: -45.0 to 45.0; KP: 0.0 to 500; KD: 0.0 to 5.0; Torque: -18.0 to 18.0

- **CAN ID** section: Supports reading and customizing the ID. Modified CAN IDs take effect after a power cycle.
- **Motor Mode** section: Supports Enable, Save zero, and Disable. Click the corresponding buttons.
- After reading the ID, switch or adjust the control mode (MIT/Position/Speed). The pre-send command is displayed in real-time in the **CAN Frame** section.
- After clicking **Send**, the feedback frame status is parsed and displayed in the **CAN Frame** section.
- **Transmit** section: Set an interval and check the box to send commands continuously. Uncheck to stop.
- When **Periodic send** is selected, data on the **CAN Frame** section and **Telemetry** page updates in real-time.

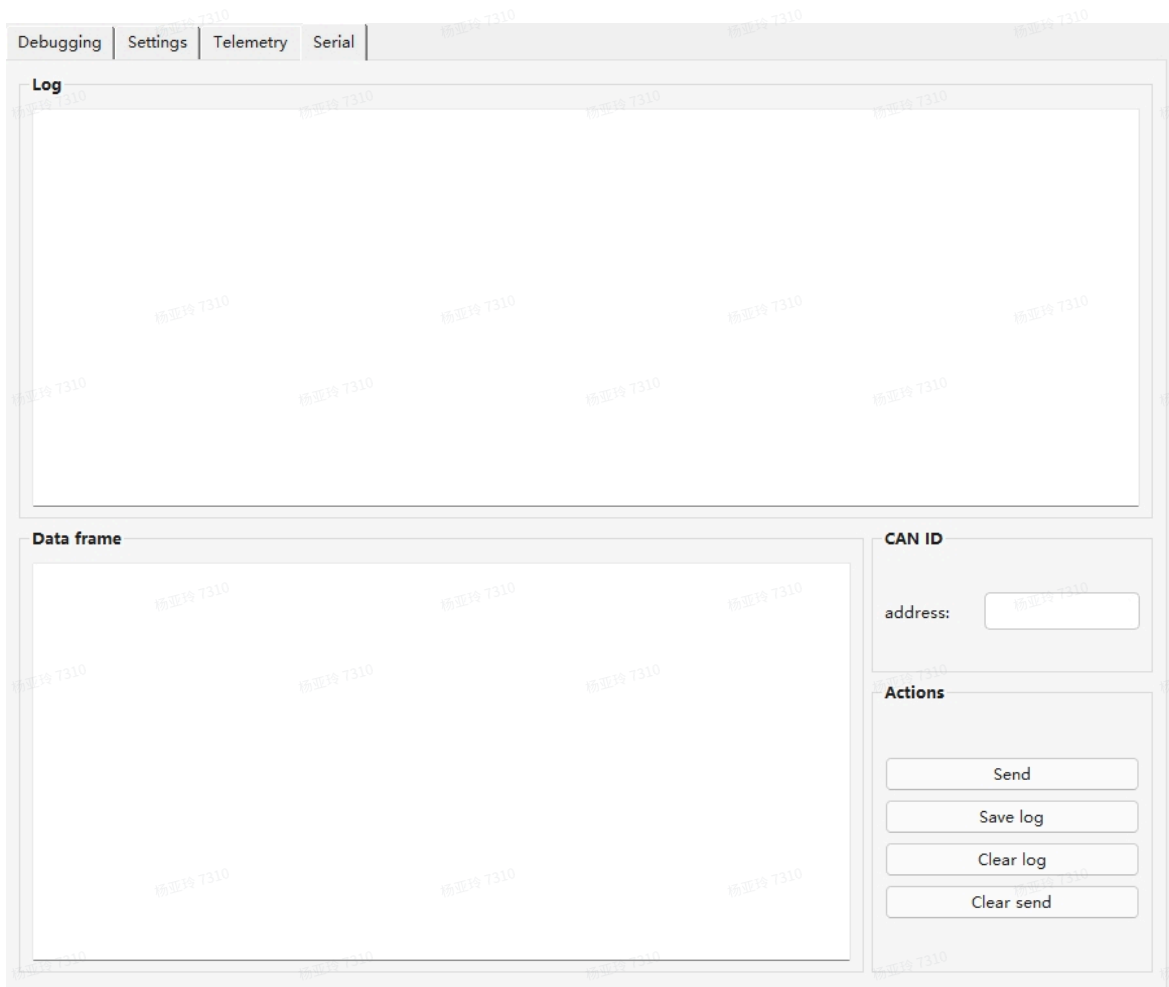
1.5 Parameter Settings



The screenshot shows the 'Settings' tab of a control interface. It features a 'Control Settings' section with four input fields: 'Speed KP:', 'Speed KI:', 'Position KP:', and 'Position KI:'. Below these fields are two buttons, 'Read' and 'Write'.

- Currently only supports reading and setting the parameters as shown above. You can first read the default parameters, then fine-tune and write them back.

1.6 Serial Port Debug



The screenshot shows the 'Serial' tab of a control interface. It features a 'Log' section with a large text area for displaying log messages. Below the log is a 'Data frame' section with a large text area for entering data. To the right of the data frame is a 'CAN ID' section with an 'address:' input field. Below the CAN ID section is an 'Actions' section with four buttons: 'Send', 'Save log', 'Clear log', and 'Clear send'.

- All commands sent during testing are displayed in the log area.
- Supports manual debugging, requiring you to write the CAN ID and data (in hexadecimal format).